

Investigating the Roles of MAPKs as Precision Oncology Targets: A Multi-cancer Expression and Survival Study

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Abstract

Background/Aim: Cancer remains a leading cause of mortality globally, driven by complex molecular mechanisms and characterized by significant biological heterogeneity across cancer types. We aimed to discover mitogen-activated protein kinases (MAPKs) family members as both biomarkers and therapeutic targets in different cancer types. MAPKs are key signaling molecules regulating cell proliferation, differentiation, stress response, and apoptosis. Dysregulation of MAPK pathways has been implicated in the onset and progression of multiple cancers, contributing to tumor growth, metastasis, and therapeutic resistance. Given their diverse roles across cancer types, systematic analysis of MAPK gene expression, mutations, and interactions with tumor microenvironment is essential.

Materials and Methods: The present study undertakes a comprehensive transcriptomic analysis of MAPKs in nine major cancer types using RNA-Seq datasets from The Cancer Genome Atlas (TCGA). RNA-Seq data were analyzed to identify differentially expressed MAPKs across nine cancer types using DESeq2, NOIseq, and limma. Significant genes (adjusted $p < 0.05$) were subjected to GO and KEGG enrichment (EnrichR), mutational profiling (cBioPortal), and Kaplan-Meier survival analysis. Tumor immune infiltration was further assessed using TIMER2.0 to explore immune-gene interactions.

Results: Through rigorous differential expression analysis, we identified key MAPKs that are significantly altered in liver hepatocellular carcinoma (LIHC) and lung adenocarcinoma (LUAD). Specifically, six MAPKs (MAPK3, MAPK7, MAPK9, MAPK10, MAPK12, and MAPK13) were found to be differentially expressed in LIHC, while MAPK6 emerged as the sole significant candidate in LUAD. Functional enrichment and pathway analysis revealed that these MAPKs are involved in critical oncogenic pathways, including MAPK-mediated transcriptional regulation and stress-activated signaling cascades. Mutational profiling and survival analysis further validated the prognostic significance of these

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genes, with several MAPKs showing strong associations with reduced patient survival. Tumor immune infiltration analysis indicated potential roles of these MAPKs in modulating immune responses within the tumor microenvironment. Also, MAPKs identified in this study are structurally related, suggesting that targeting them collectively may enhance therapeutic efficacy and overcome resistance mechanisms.

Conclusion: Our integrated approach underscores the value of MAPK family members as both biomarkers and therapeutic targets in LIHC and LUAD. This study contributes important insights into MAPK-related oncogenic processes and supports the development of targeted therapies under the framework of precision oncology.

Keywords: Mitogen-activated protein kinases, MAPKs, cancers, biomarkers, differential expression analysis, survival analysis.

Introduction

Cancer remains one of the most pressing global health challenges of the 21st century, significantly contributing to public health concerns, social burden, and economic impact. It accounts for about one in six deaths worldwide and nearly one-quarter of all fatalities caused by noncommunicable diseases (NCDs) (1). Among individuals aged 30 to 69 years, cancer is responsible for roughly 30% of premature deaths from NCDs and is among the top three causes of death in the majority of countries. In 2022, lung cancer emerged as both the most commonly diagnosed and deadliest cancer globally, with an estimated 2.5 million new cases and over 1.8 million deaths. It contributed to 12.4% of all new cancer cases and 18.7% of cancer-related deaths. Female breast cancer was the second most common, with 2.3 million new cases (11.6%), and was the fourth leading cause of cancer deaths globally, responsible for about 666,000 deaths. Among women, it remained the most diagnosed and the top cause of cancer-related mortality. In men, prostate cancer ranked as the second most frequently diagnosed cancer, with 1.5 million new cases and nearly 397,000 deaths, making it the fifth leading cause of cancer death (2).

Liver cancer ranked as the sixth most commonly diagnosed cancer worldwide in 2022, with an estimated 865,000 new cases reported (3). Liver cancer is the second leading cause of cancer-related deaths among men globally, with incidence and mortality rates generally two

to three times higher in men than in women across most regions. Hepatocellular carcinoma accounts for approximately 75% to 85% of all primary liver cancer cases, while intrahepatic cholangiocarcinoma comprises around 10% to 15%. Chronic infections with hepatitis B virus (HBV) or hepatitis C virus (HCV) are major risk factors and are associated with 21% to 55% of HCC cases worldwide. (4). Additional risk factors for HCC include exposure to aflatoxins, excessive alcohol intake, obesity, type 2 diabetes, and tobacco use (5). HNSC is the seventh most prevalent cancer globally, contributing to more than 660,000 new cases and around 325,000 deaths each year (6). These cancers pose a major global health challenge, with patients commonly enduring substantial hardships due to both the progression of the disease and the adverse effects of its treatment (7).

The RAS protein plays a key role in regulating cell proliferation and survival in response to growth factors, in both normal and cancerous cells (8). Once activated, RAS triggers multiple downstream signaling pathways, with the most well-studied being the RAF/MEK/ERK cascade (also known as the classical MAPK pathway), the phosphoinositide 3-kinase (PI3K) pathway, and the Ral-GEF signaling route (9). Mitogen-activated protein kinases (MAPKs) are a group of protein kinases with highly conserved functions across evolution, from simple organisms like yeast to complex species such as humans. These kinases play a critical role by phosphorylating specific serine and threonine residues on target proteins,

thereby influencing a wide array of cellular processes, including gene regulation, cell division, motility, metabolism, and apoptosis (10-15).

In multicellular organisms, three major subfamilies of MAPKs have been extensively studied. These include the extracellular signal-regulated kinases (ERK1 and ERK2), the c-Jun N-terminal kinases (JNK1, JNK2, and JNK3), and the p38 MAPK family, which consists of four isoforms: p38 α , p38 β , p38 γ , and p38 δ (16). Multiple approaches are currently under investigation to inhibit MAPK signaling as a strategy for cancer treatment. Although direct inhibitors of RAS have not yet demonstrated clinical success, alternative therapeutics have shown promising efficacy. These include targeting synthetic lethal partners specific to RAS-mutant tumors and inhibiting key downstream components of the pathway, such as RAF and MEK (17).

Precision oncology, often referred to as biomarker-guided therapy, has rapidly advanced clinical care by improving treatment outcomes (18). This approach focuses on identifying actionable biomarkers and therapeutic targets that are linked to unfavorable prognosis and diminished survival in cancer patients, thereby offering more personalized and effective treatment strategies (19). Recent progress in diagnostics, drug discovery, and biological research through modern techniques is poised to play a crucial role in enhancing cancer treatment within the framework of precision oncology (20). Large-scale datasets from diverse cancer types, produced through various experimental studies by research groups worldwide, are freely accessible through resources such as The Cancer Genome Atlas (TCGA) and the Genomic Data Commons (GDC) portal (21). The genomic data available in these repositories are increasingly being leveraged to enable precisely targeted therapies in cancer research. Numerous studies have drawn on these resources to investigate the genetic underpinnings of various cancers, revealing significant disruptions in multiple genes (22).

In this study, nine cancer types, breast invasive carcinoma (BRCA), lung squamous cell carcinoma (LUSC), lung adenocarcinoma (LUAD), liver hepatocellular carcinoma (LIHC), head and neck squamous cell

carcinoma (HNSC), kidney renal clear cell carcinoma (KIRC), kidney renal papillary cell carcinoma (KIRP), stomach adenocarcinoma (STAD), and prostate adenocarcinoma (PRAD) were selected, as they collectively represent a substantial share of the global cancer burden. Each of these cancers has distinct biological behaviors, risk factors, and clinical challenges, making them critical targets for molecular studies aimed at uncovering disease-specific and shared mechanisms that could inform better therapeutic strategies. We adopted an unbiased and comprehensive approach to investigate the biological heterogeneity and genomic landscape of cancer. By systematically analyzing pan-cancer samples, we aimed to identify novel biomarkers and druggable targets relevant to diagnosis, prognosis, and therapy. Differential expression analysis was conducted to examine the expression patterns of MAPKs across these cancer types. This integrative approach facilitated the discovery of potential molecular markers and therapeutic targets.

Furthermore, we assessed the alteration frequencies and performed survival analysis in patients with genetic modifications in the identified MAPKs. Overall, this study offers valuable insights into the differential expression of MAPKs across nine cancer types, contributing to the genomic understanding and therapeutic exploration within the cancer research community.

Materials and Methods

RNA-Seq data. Normal and tumor samples from nine distinct cancer types were obtained from the TCGA data portal (<https://portal.gdc.cancer.gov/>). Only site-specific samples corresponding to each cancer type were included, while non-specific or unclassified samples were excluded from the analysis. To extract mRNA expression data (HTSeq-counts generated using the Illumina HiSeq platform), the UCSC Xena browser (<https://xenabrowser.net/>) was utilized. Additionally, associated phenotypic information, including age at diagnosis, weight, batch effects, and gender was retrieved from the Xena platform.

Differential gene expression. The collected data were organized in an Excel sheet, where each row represented an expressed RNA and each column corresponded to either tumor or control samples. Differential expression analysis was carried out using R programming, utilizing a separate sheet containing sample type annotations and sample count information to identify differentially expressed genes (DEGs). The DESeq2, NOIseq, and limma packages were used for pre-processing (Normalization, log2 transformation, and batch correction) and DEG identification, respectively. Significantly differentially expressed MAPKs were identified based on log₂ fold change and the adjusted *p*-value criteria. A log₂ fold change of zero was considered the threshold, with negative values indicating down-regulated genes and positive values indicating up-regulated genes. Genes with an adjusted *p*-value of less than 0.05 were considered statistically significant.

GO term and pathway enrichment analyses. Gene Ontology (GO) term and pathway analysis are essential bioinformatics tools used to interpret high-throughput genomic or transcriptomic data (like RNA-seq or microarray). They help identify biological meaning in differentially expressed genes. GO helps understand what functions are overrepresented in the gene list, provides functional annotation of genes based on shared biological characteristics, facilitates hypothesis generation, and is helpful in identifying key regulatory processes and understanding cellular roles of genes. Pathway enrichment analysis identifies biological pathways (*e.g.*, KEGG, Reactome, WikiPathways) enriched in a set of genes. It helps uncover biological mechanisms and molecular interactions involved in the condition of interest, enables system-level understanding of disease, treatment response, or cellular state and assists in identifying potential drug targets or biomarkers (23).

GO analysis was conducted to uncover key biological characteristics associated with the RNA-seq data. Up-regulated and down-regulated genes were analyzed separately using the enrichR tool for GO enrichment. The

resulting gene sets were further categorized into functional groups based on GO terms related to Biological Process (BP), Molecular Function (MF), and Cellular Component (CC).

The top 3 significantly up-regulated and down-regulated GO categories were considered. These significantly enriched terms could help us a lot to understand further the role that DEGs play in cancers. Pathway enrichment analysis for the significant MAPKs was performed using the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways available through the Enrichr database. GO terms and pathways with a *p*-value less than 0.05 were considered statistically significant.

Mutational analysis. Mutational (alteration frequency) analysis is a powerful method used to assess the frequency at which specific genes are genetically altered across different cancer types. Alteration frequency refers to how frequently a gene shows genomic alterations such as mutations, copy number variations (amplifications/deletions), or mRNA expression changes across patient samples of a cancer type (24). Genes with high alteration frequency may play key roles in tumorigenesis and can be prioritized for further functional or therapeutic studies. Frequently amplified or overexpressed genes may act as oncogenes, and frequently deleted or mutated genes may function as tumor suppressors (25). Hence, the altered genes can serve as diagnostic, prognostic, or predictive biomarkers. Mutational frequencies of the identified candidate genes were examined using TCGA datasets accessed via cBioPortal (<https://www.cbioportal.org/>). Each mutation was mapped onto the two-dimensional structures of the corresponding proteins, and their occurrence frequencies were recorded (22).

Survival analysis. Survival analysis traditionally refers to the statistical examination of the time until death within a group of individuals. More broadly, it involves the analysis of time-to-event data, which measures the duration from a defined starting point to a specific event of interest. For example, individuals might be followed

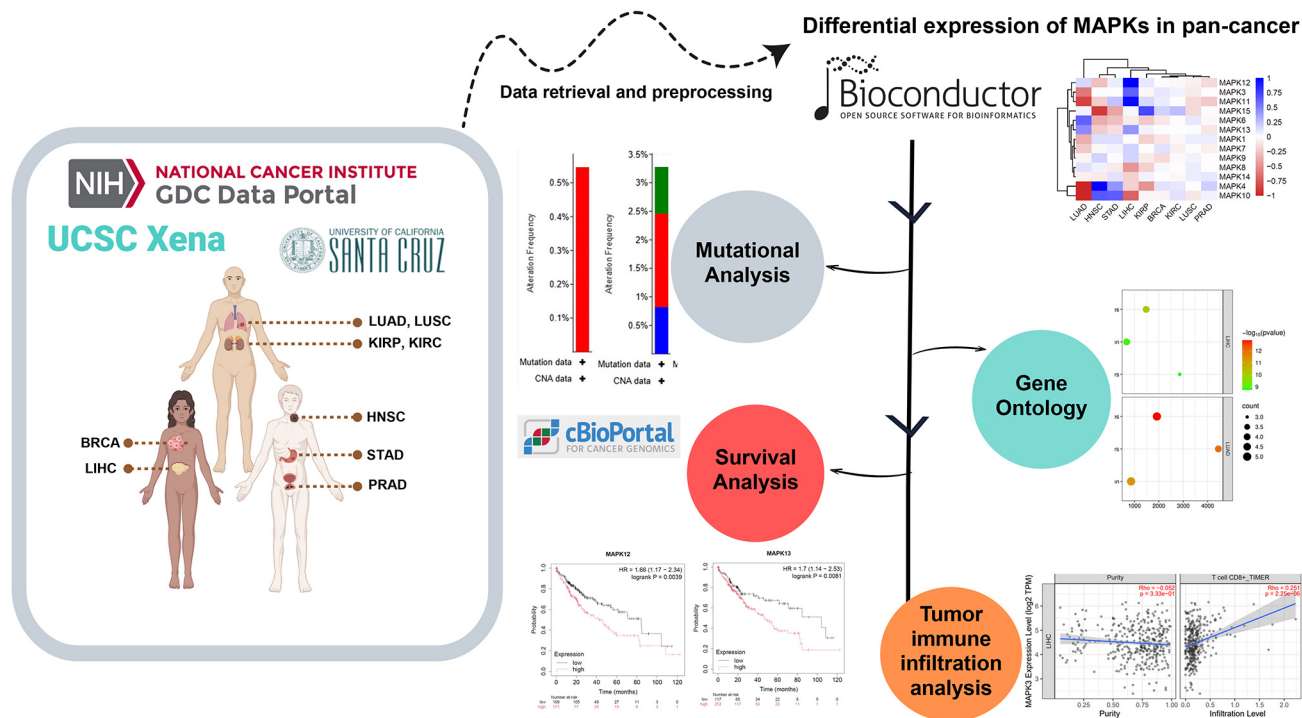


Figure 1. Graphical flowchart of the methodology, depicting data acquisition, differential gene expression analysis, and subsequent mutational, enrichment, survival, and tumor infiltration analyses of the selected mitogen-activated protein kinases genes (MAPKs).

from birth to the onset of some disease, or the survival time after the diagnosis of some disease might be studied (26). Survival analysis methods are typically applied to data collected prospectively over time, such as those obtained from cohort studies or clinical trials (27). To assess the impact of alterations in the identified candidate MAPKs on overall survival across the previously mentioned cancer types, survival analysis was conducted using TCGA data accessed via cBioPortal. Kaplan-Meier (KM) survival curves were generated, and statistical significance was evaluated using the log-rank test p -value (28).

Tumor immune infiltration analysis. Tumor-infiltrating immune cells play a critical role in determining treatment response and patient prognosis in cancer. The diversity of immune cell populations within the tumor micro-environment contributes to tumor heterogeneity,

presenting both opportunities and challenges in understanding the dynamic interplay between immune and cancer cells (29). Cancer immunotherapy is designed to enhance the activity of cytotoxic T lymphocytes (CTLs) within tumors, support the priming of tumor-specific CTLs in lymphoid tissues, and establish robust and lasting antitumor immunity. During the priming phase, CD4⁺T cells provide essential help signals to CD8⁺ T cells through specialized dendritic cells, thereby enhancing the strength and quality of the CTL response (30). RNA-seq-based gene expression profiling is a well-established method for tumor characterization and has been extensively applied to generate comprehensive transcriptomic datasets across various cancer types. However, while these data provide valuable molecular insights, they do not directly reveal the composition of immune cell populations within tumors, necessitating computational approaches for inference.

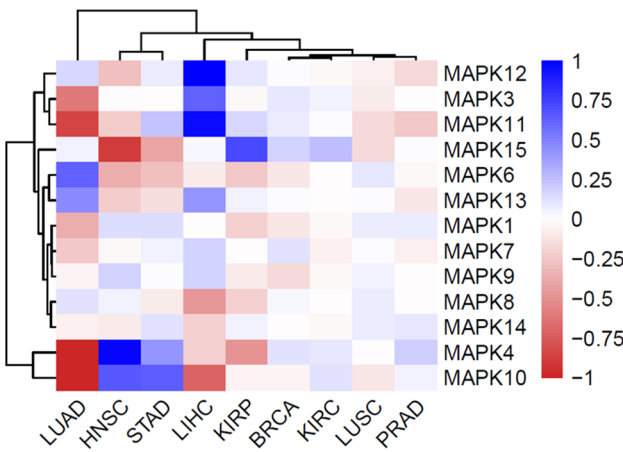


Figure 2. Heatmap showing differentially expressed mitogen-activated protein kinases (MAPKs) across nine cancer types.

TIMER2.0 is a publicly accessible web server developed for the research community, built using the Shiny web framework for R. It comprises three main components: Immune, Exploration, and Estimation. The Immune component includes four modules that enable users to examine associations between estimated immune cell infiltration and various factors such as gene expression, somatic mutations, somatic copy number alterations (sCNAs), and clinical outcomes across TCGA cancer cohorts (31).

Results

The flowchart of the study is depicted in Figure 1. After averaging the expression values for duplicate gene entries, a final set of unique genes was retained. The processed mRNA datasets were then subjected to differential expression analysis using the limma package. Based on the defined thresholds (p -value < 0.05 and $|\log_2(\text{fold change})| > 0$) for up-regulated and $|\log_2(\text{fold change})| < 0$ for down-regulated, a set of differentially expressed genes (DEGs) was identified for all the above cancer types following the same procedure for each cancer type individually. After obtaining the differentially expressed genes for each cancer type, we extracted only the MAPKs for each cancer type. In Figure 2, the heatmap shows the expression level

Table I. Sample size of nine different cancer types.

S. No.	Cancer type	No. of samples	
		Tumor	Normal
1.	BRCA	844	69
2.	HNSC	200	05
3.	KIRC	238	161
4.	KIRP	217	15
5.	LIHC	366	47
6.	LUAD	433	11
7.	LUSC	453	38
8.	PRAD	459	50
9.	STAD	261	21

BRCA: Breast invasive carcinoma; HNSC: head and neck squamous cell carcinoma; KIRC: kidney renal clear cell carcinoma; KIRP: kidney renal papillary cell carcinoma; LIHC: liver hepatocellular carcinoma; LUAD: lung adenocarcinoma; LUSC: lung squamous cell carcinoma; PRAD: prostate adenocarcinoma; STAD: stomach adenocarcinoma.

of each MAPK on the basis of logFC value. Blue indicates the up-regulated genes and red are the indicative of down-regulated. The sample sizes for each cancer type are listed in Table I.

After that, we analyzed the shortlisted MAPKs for each cancer type individually, based on both p -value and log fold change (logFC). MAPKs that were significantly up-regulated (p -value < 0.05 , positive logFC) or down-regulated (p -value < 0.05 , negative logFC) were then examined in relation to survival trends. Finally, MAPKs with logFC values that followed the same trend as observed in the survival analysis were selected. Hence, after analyzing each cancer types we got final sets of MAPKs in LIHC and LUAD and no significant MAPKs were found in other cancer types.

MAPK3 with (p -value $1.24\text{E-}32$ and logFC 0.624897729), MAPK7 (p -value 0.001095477 and logFC 0.191065754), MAPK9 (p -value 0.000798419 and logFC 0.175540836), MAPK10 (p -value $3.88\text{E-}15$ and logFC -0.70465399), MAPK12 (p -value $2.05\text{E-}19$ and logFC 1.29189209) and MAPK13 (p -value 0.006584904 and logFC 0.416415897) are the significant genes found in LIHC and the only one gene that is MAPK6 with (p -value 0.0000000727 and logFC 0.625150634) was found to be significant in LUAD and hence chosen for further analysis.

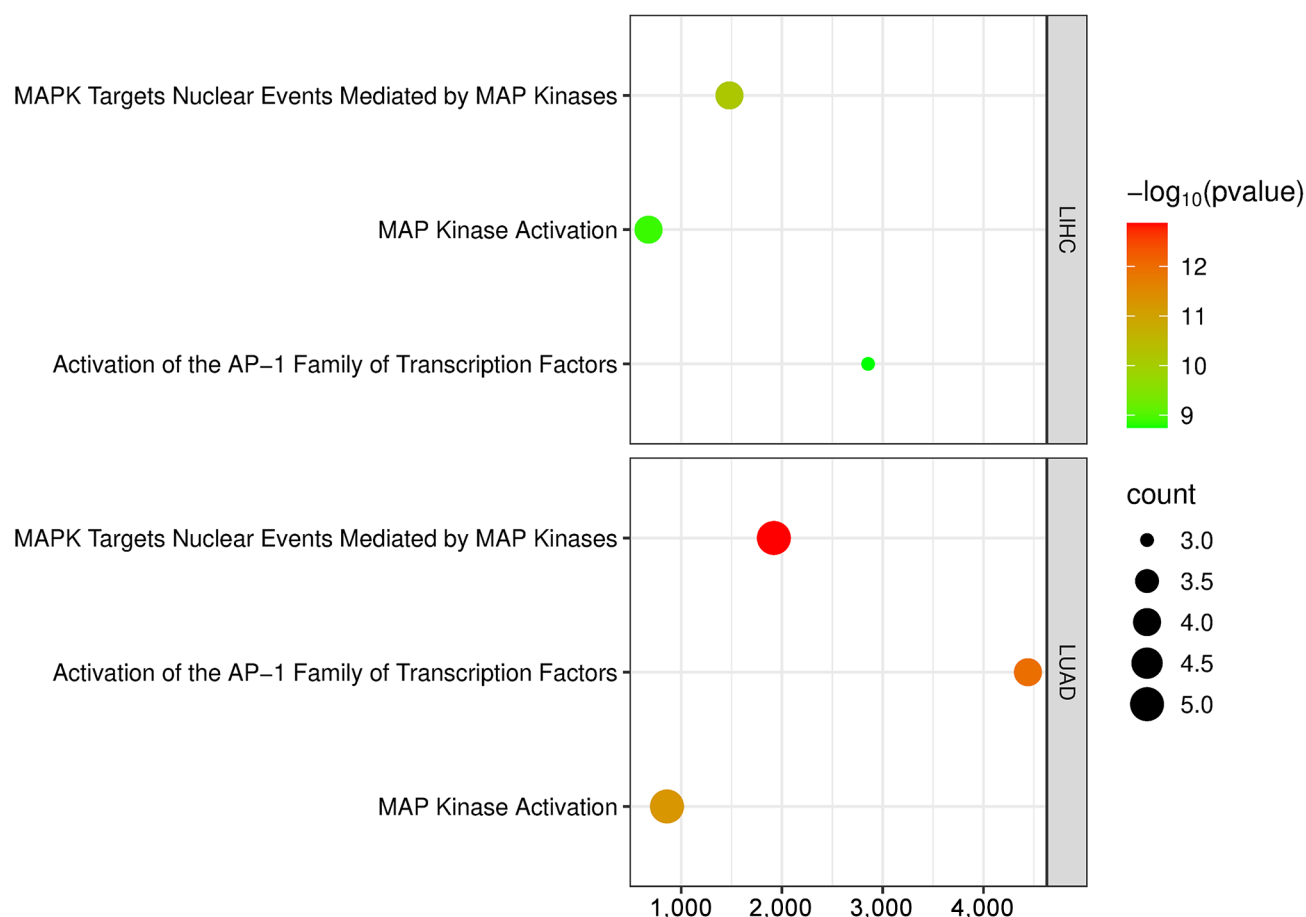


Figure 3. Top three pathways involved in liver hepatocellular carcinoma (LIHC) and lung adenocarcinoma (LUAD).

GO term and pathway enrichment analyses of selected MAPKs in LIHC and LUAD. Among 6 key MAPKs in LIHC, MAPK3, MAPK9 and MAPK10 participated in all the top three significant pathways with MAPK targets nuclear events mediated by MAP kinases (p -value=7.06E-11) being the most significant. The association of MAPKs with three significant pathways is illustrated in Figure 3. MAPK targets nuclear events mediated by MAP kinases is a pathway that includes the nuclear transcriptional regulation events triggered by MAPK signaling. MAPKs (especially ERK1/2) phosphorylate transcription factors in the nucleus (*e.g.*, Elk-1, Myc, ATF), promoting expression of genes that regulate, cell proliferation, survival, differentiation (32).

In cancer, overactive MAPK signaling leads to persistent nuclear activation, driving uncontrolled cell cycle progression and resistance to apoptosis. The MAP kinase activation pathway represents the initial activation steps of the MAPK signaling cascade. Involves RAS-RAF-MEK-ERK and JNK/p38 pathways. Mutations or amplifications in upstream regulators such as RAS and BRAF are common oncogenic events in cancers. Aberrant activation contributes to oncogenic transformation, tumor progression, and resistance to targeted therapies (33).

Activation of the AP-1 family of transcription factors, a pathway composed of proteins like c-Jun, c-Fos, is activated downstream of MAPKs, controls the expression of genes involved in cell cycle regulation, migration and

Table II. Reactome pathway in liver hepatocellular carcinoma.

S. No.	Pathway	p-Value	Genes
1.	MAPK targets nuclear events mediated by MAP kinases	7.06E-11	MAPK10, MAPK9, MAPK7, MAPK3
2.	MAP kinase activation	1.33E-09	MAPK10, MAPK9, MAPK7, MAPK3
3.	Activation of the AP-1 family of transcription factors	1.80E-09	MAPK10, MAPK9, MAPK3

MAPK: Mitogen-activated protein kinase.

Table III. Reactome pathway in liver hepatocellular carcinoma.

S. No.	Pathway	p-Value	Genes
1.	MAPK targets nuclear events mediated by MAP kinases	1.34E-13	MAPK10, MAPK9, MAPK11, MAPK8, MAPK7
2.	Activation of the AP-1 family of transcription factors	1.10E-12	MAPK10, MAPK9, MAPK11, MAPK8
3.	MAP kinase activation	5.51E-12	MAPK10, MAPK9, MAPK11, MAPK8, MAPK7

MAPK: Mitogen-activated protein kinase.

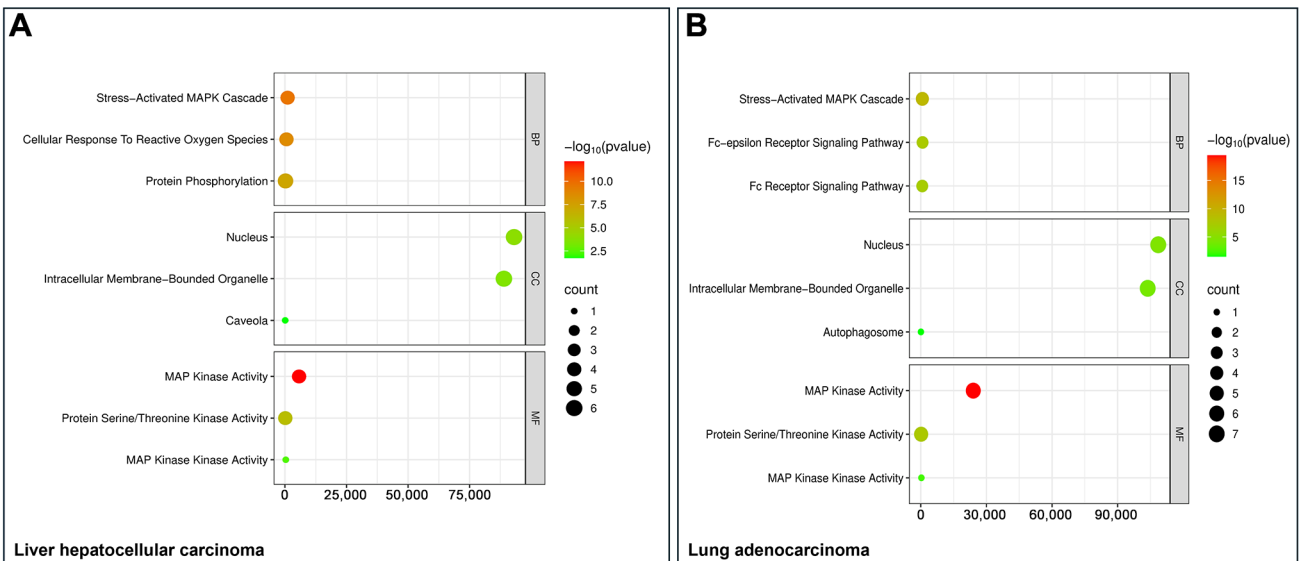


Figure 4. Gene Ontology (GO) enrichment bubble plot, (A) for liver hepatocellular carcinoma (LIHC) and (B) for lung adenocarcinoma (LUAD). p-Values are represented by colors, and gene counts are represented by bubble size.

invasion, and inflammation [34]. MAPK12 and MAPK13 were not present in the top three pathways within the above-mentioned significant threshold (i.e., $p\text{-value} < 0.05$). Similarly, the involvement of MAPK6 was evaluated in the top three significantly enriched pathways in LUAD. Although MAPK6 was not represented among the top three pathways, its participation became evident upon extending the analysis. However, these pathways were

present in the top three of the threshold values in both cancer types, and these are very crucial for cancer studies. The top three significant pathways for LIHC and LUAD, respectively, are illustrated in Table II and Table III.

Also, the top three BP, MF, and CC terms were screened based on their significant $p\text{-value}$. Figure 4A-B shows top 3 significant GO-BP, MF, and CC. The most significant GO-BP, MF, and CC terms were stress-activated MAPK cascade

Table IV. *Gene Ontology-Biological Process in liver hepatocellular carcinoma.*

S. No.	Term	p-Value	Genes
1.	Stress-activated MAPK cascade	2.51E-10	MAPK10, MAPK9, MAPK3, MAPK13
2.	Cellular response to reactive oxygen species	2.44E-09	MAPK9, MAPK7, MAPK3, MAPK13
3.	Protein phosphorylation	5.63E-08	MAPK10, MAPK9, MAPK12, MAPK3, MAPK13

MAPK: Mitogen-activated protein kinase.

Table V. *Gene Ontology-Cellular Component in liver hepatocellular carcinoma.*

S. No.	Term	p-Value	Genes
1.	Nucleus	1.27E-04	MAPK10, MAPK9, MAPK7, MAPK12, MAPK3, MAPK13
2.	Intracellular membrane-bounded organelle	2.99E-04	MAPK10, MAPK9, MAPK7, MAPK12, MAPK3, MAPK13
3.	Caveola	0.018459	MAPK3

MAPK: Mitogen-activated protein kinase.

Table VI. *Gene Ontology-Molecular Function in liver hepatocellular carcinoma.*

S. No.	Term	p-Value	Genes
1.	MAP kinase activity	7.42E-13	MAPK10, MAPK9, MAPK3, MAPK13
2.	Protein serine/threonine kinase activity	1.23E-06	MAPK7, MAPK12, MAPK3, MAPK13
3.	MAP kinase kinase activity	4.19E-03	MAPK10

MAPK: Mitogen-activated protein kinase.

Table VII. *Gene Ontology-Biological Process in lung adenocarcinoma.*

S. No.	Term	p-Value	Genes
1.	Stress-activated MAPK cascade	5.85E-10	MAPK10, MAPK9, MAPK11, MAPK8
2.	Fc-epsilon receptor signaling pathway	4.03E-08	MAPK10, MAPK9, MAPK8
3.	Fc receptor signaling pathway	6.80E-08	MAPK10, MAPK9, MAPK8

MAPK: Mitogen-activated protein kinase.

(p -value=2.51E-10), MAP kinase activity (p -value=7.42E-13), and nucleus (p -value=1.27E-04). MAPK10, MAPK9, MAPK3, MAPK7, MAPK12 and MAPK13 were a part of common GO-BP terms. Similarly, in the case of MAPK6, nucleus and intracellular membrane-bounded organelle were the enriched Gene Ontology Cellular Component (GO-CC) terms, while MAP kinase activity and protein serine/threonine kinase activity were the key functions identified under Gene Ontology Molecular Function (GO-MF). These ontology terms are considered crucial in

cancer progression (Table IV, Table V, Table VI, Table VII, Table VIII and Table IX).

Mutational frequencies. While analyzing TCGA datasets through cBioPortal, mutations were identified and mapped along with their frequencies across various cancer types (35). Mutation frequencies in MAPK3, MAPK7, MAPK9, MAPK10, MAPK12, and MAPK13 were estimated to be 0.5%, 3%, 2.5%, 1.4%, 1.4%, and 3% respectively in case of LIHC (Figure 5A). Similarly, the

Table VIII. Gene Ontology-Cellular Component in lung adenocarcinoma.

S. No.	Term	p-Value	Genes
1.	Nucleus	2.85E-05	MAPK10, MAPK9, MAPK11, MAPK8, MAPK7, MAPK6, MAPK15
2.	Intracellular membrane-bounded organelle	7.74E-05	MAPK10, MAPK9, MAPK11, MAPK8, MAPK7, MAPK6, MAPK15
3.	Autophagosome	0.028695	MAPK15

MAPK: Mitogen-activated protein kinase.

Table IX. Gene Ontology-Molecular Function in lung adenocarcinoma.

S. No.	Term	p-Value	Genes
1.	MAP kinase activity	3.64E-20	MAPK10, MAPK9, MAPK11, MAPK8, MAPK6, MAPK15
2.	Protein serine/threonine kinase activity	2.90E-08	MAPK11, MAPK8, MAPK7, MAPK6, MAPK15
3.	MAP kinase kinase activity	0.00489	MAPK10

MAPK: Mitogen-activated protein kinase.

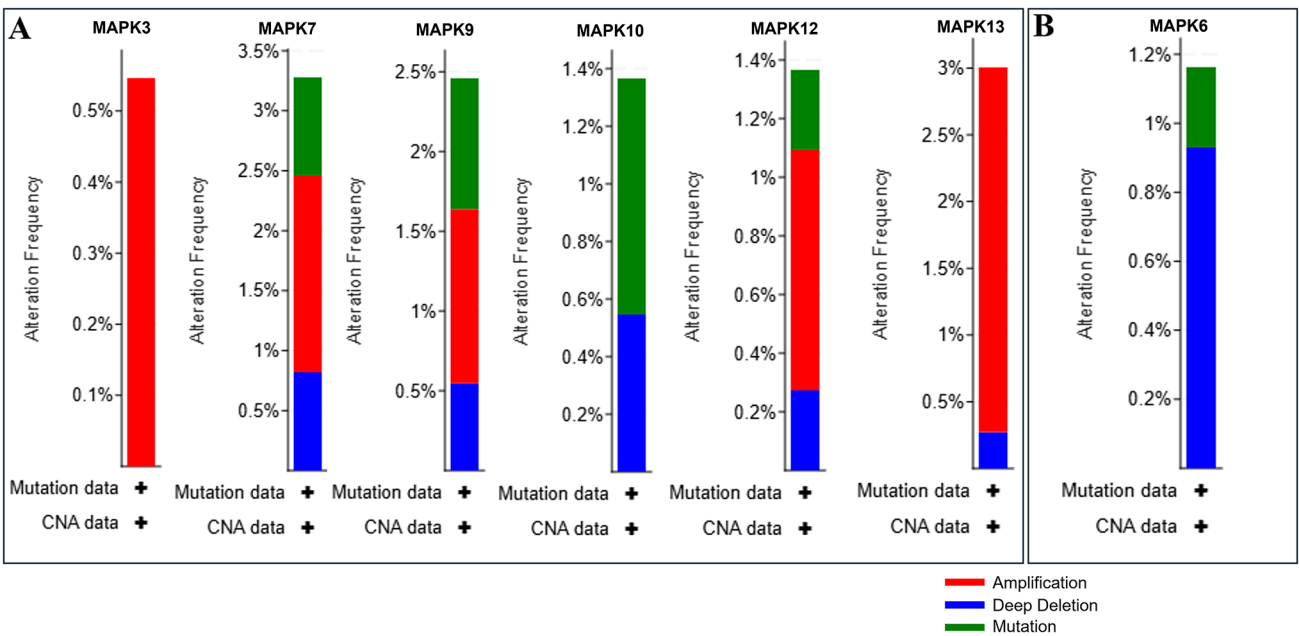


Figure 5. Frequency of point mutations and their types for (A) mitogen-activated protein kinase (MAPK) 3, MAPK7, MAPK9, MAPK10, MAPK12, and MAPK13 in liver hepatocellular carcinoma (LIHC) and (B) for MAPK6 in lung adenocarcinoma (LUAD). The figure was generated using cBioPortal, based on data retrieved from The Cancer Genome Atlas (TCGA) datasets.

alteration frequency of MAPK6, the only significant gene in LUAD, was found to be 1.2% (Figure 5B).

These mutations may contribute to structural alterations in the MAPKs, potentially resulting in their

dysfunction and promoting the progression of liver and lung cancers. We have generated an OncoPrint showing the genomic alterations in LIHC and LUAD and elucidated biomarkers, MAPK3, MAPK7, MAPK9, MAPK10, MAPK12,

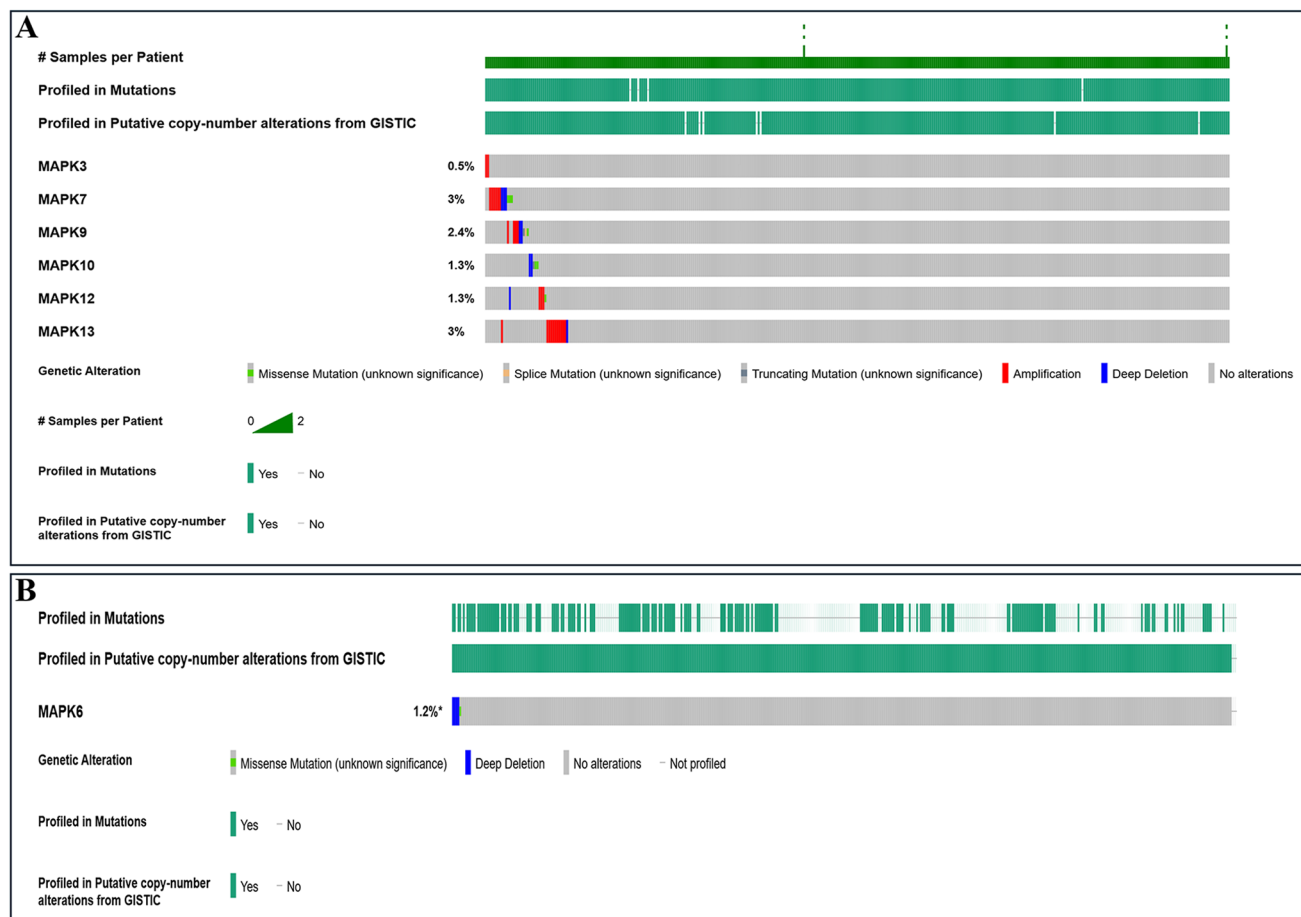


Figure 6. OncoPrint showing the alteration frequency of (A) mitogen-activated protein kinase (MAPK) 3, MAPK7, MAPK9, MAPK10, MAPK12, and MAPK13 in liver hepatocellular carcinoma (LIHC) and (B) MAPK6 in lung adenocarcinoma (LUAD). The figure was generated through the cBioPortal based on The Cancer Genome Atlas (TCGA) datasets.

and MAPK13, and MAPK6, respectively (Figure 6A-B). Moreover, in LIHC, MAPK7 and MAPK13 showed the highest percent of alteration frequency, which is 3%, making them suitable biomarkers for LIHC.

Survival of patients. The Kaplan-Meier survival analyses reveal that higher expression levels of MAPK3, MAPK7, MAPK9, MAPK12, and MAPK13 are consistently associated with reduced patient survival times and the higher expression of MAPK10 is associated with increased survival times. Specifically, patients exhibiting high MAPK3 expression showed significantly reduced survival probabilities, with a hazard ratio (HR) of 1.91 and a highly

significant p -value of 0.00018, indicating nearly twice the risk of death compared to those with low expression. Similarly, high MAPK7 expression mirrored this trend, with an identical HR of 1.91 and the same level of statistical significance, suggesting a robust and critical role of MAPK7 in disease progression. MAPK9 also demonstrated a notable negative impact on survival, with an HR of 1.6 and a p -value of 0.0068, highlighting that elevated MAPK9 expression moderately but significantly increases the risk of mortality (Table X).

In the case of MAPK12, high expression levels were associated with a 1.66-fold increased risk of death (p -value=0.0039), underscoring its potential role as a poor

Table X. Survival pattern of identified MAPKs in LIHC and LUAD.

S. No.	Genes	Cancer type	Low expression cohort (months)	High expression cohort (months)	Hazard ratio (HR)	95% CI
1.	MAPK3	LIHC	70.53	38.23	1.91	1.35-2.7
2.	MAPK7	LIHC	71.03	33.5	1.91	1.35-2.7
3.	MAPK9	LIHC	59.7	37.83	1.6	1.13-2.26
4.	MAPK10	LIHC	52	70.53	0.67	0.47-0.97
5.	MAPK12	LIHC	84.4	42.37	1.66	1.17-2.34
6.	MAPK13	LIHC	104.17	47.43	1.7	1.14-2.53
7.	MAPK6	LUAD	54.4	34.87	1.78	1.31-2.41

MAPK: Mitogen-activated protein kinase; LIHC: liver hepatocellular carcinoma; LUAD: lung adenocarcinoma; CI: confidence interval.

prognostic marker. Lastly, high MAPK13 expression correlated with a significantly poorer survival probability (HR=1.7, p -value=0.0081), further emphasizing its unfavorable prognostic value. Similarly, the high expression of MAPK6 leads to the poor survival of LUAD patient with (HR=1.78, p -value=0.0002) as shown in Figure 7B. Collectively, these results suggest that overexpression of these MAPK genes consistently predicts worse survival across patients, implying their potential as critical biomarkers for prognosis and possibly as therapeutic targets to improve clinical outcomes in cancer management (Figure 7A-B and Table XI).

Tumor immune infiltration analysis. TIMER 2.0 is an online platform designed for the systematic analysis of immune cell infiltration across various cancer types. In this study, TIMER 2.0 was used to evaluate the correlation between the expression of identified MAPKs and tumor-infiltrating immune cells including CD4⁺ T cells, CD8⁺ T cells, NK cells, macrophages, B cells, and dendritic cells in patients with LIHC and LUAD. Spearman correlation analysis was performed to determine statistical significance, and gene expression levels were represented as RSEM values (36). MAPKs expression levels were significantly correlated with tumor purity and the infiltrating levels of (CD4⁺, CD8⁺, T cell NK, macrophage, B cell, and dendritic cell) in LIHC and LUAD patients.

Table XI shows significant correlations of MAPKs mRNA expression with CD4⁺, CD8⁺, T cell NK, macrophage, B cell, and dendritic cell infiltrating levels across the TCGA-LIHC

and TCGA-LUAD cohort. mRNA expression levels against tumor purity are also demonstrated in the same table. Spearman's correlation and estimated statistical significance are displayed for each scatter plot. Scatterplots showing significant correlations of MAPKs mRNA expression with tumor infiltrates across the LIHC and LUAD cohorts are included in Figure 8 and Figure 9, respectively.

Discussion

Our study aimed to identify significant MAPK family members across nine cancer types LUAD, LIHC, LUSC, BRCA, KIRC, KIRP, HNSC, PRAD, and STAD using RNA-seq data from TCGA, with significance assessed based on threshold p -values, log fold change and survival correlations, *etc.* Out of the 13 known MAPKs, only seven (MAPK3, MAPK6, MAPK7, MAPK9, MAPK12, and MAPK13) emerged as significantly up-regulated, and MAPK10 as significantly down-regulated and notably, their significance was confined to LIHC and LUAD. Although our intention was to identify MAPKs broadly relevant across all these cancer types, statistical and survival filtering limited our findings to these two cancers. This does not preclude the importance of other MAPKs as well as the cancer types but highlights the stringency of our selection criteria. These MAPKs are known for their roles in oncogenic processes MAPK3 (ERK1), MAPK7 (ERK5), and others are implicated in promoting proliferation, survival, and disrupted differentiation (37). MAPK6 (ERK3), an atypical MAPK, is notably up-regulated in LUAD and has been suggested as an emerging oncogenic driver (38).

Table XI. Tumor infiltration analysis.

S. No.	Cancer type	MAPKs	Immune infiltrates	p-Value (purity)	Rho (purity)	p-Value (immune infiltrates)	Rho (immune infiltrates)
1.	LIHC	MAPK3	CD4 ⁺	3.33×10 ⁻¹	-0.052	8.00×10 ⁻⁵	0.211
			CD8 ⁺	3.33×10 ⁻¹	-0.052	2.25×10 ⁻⁶	0.251
			T cell NK	3.33×10 ⁻¹	-0.052	2.15×10 ⁻²	0.124
			Macrophage	3.33×10 ⁻¹	-0.052	1.49×10 ⁻¹⁵	0.412
			B cell	3.33×10 ⁻¹	-0.052	4.86×10 ⁻⁹	0.308
			Dendritic cell	3.33×10 ⁻¹	-0.052	7.82×10 ⁻²⁴	0.506
		MAPK7	CD4 ⁺	6.41×10 ⁻¹	-0.025	1.01×10 ⁻⁹	0.321
			CD8 ⁺	6.41×10 ⁻¹	-0.025	1.04×10 ⁻²	0.138
			T cell NK	-	-	-	Insignificant
			Macrophage	6.41×10 ⁻¹	-0.025	4.67×10 ⁻¹⁷	0.431
			B cell	6.41×10 ⁻¹	-0.025	2.17×10 ⁻⁹	0.315
			Dendritic cell	6.41×10 ⁻¹	-0.025	4.97×10 ⁻¹⁸	0.443
		MAPK9	CD4 ⁺	8.49×10 ⁻²	0.093	4.63×10 ⁻⁷	0.267
			CD8 ⁺	8.49×10 ⁻²	0.093	4.73×10 ⁻²	0.107
			T cell NK	-	-	-	Insignificant
			Macrophage	8.49×10 ⁻²	0.093	1.36×10 ⁻¹¹	0.354
			B cell	8.49×10 ⁻²	0.093	5.77×10 ⁻⁸	0.287
			Dendritic cell	8.49×10 ⁻²	0.093	9.57×10 ⁻¹⁰	0.322
		MAPK10	CD4 ⁺	1.40×10 ⁻¹⁰	-0.336	2.22×10 ⁻⁸	0.295
			CD8 ⁺	1.40×10 ⁻¹⁰	-0.336	2.7×10 ⁻⁴	0.195
			T cell NK	-	-	-	Insignificant
			Macrophage	1.40×10 ⁻¹⁰	-0.336	7.44×10 ⁻¹⁰	0.324
			B cell	1.40×10 ⁻¹⁰	-0.336	4.03×10 ⁻³	0.154
			Dendritic cell	1.40×10 ⁻¹⁰	-0.336	5.37×10 ⁻⁹	0.308
		MAPK12	CD4 ⁺	3.18×10 ⁻⁶	-0.247	1.86×10 ⁻⁷	0.276
			CD8 ⁺	3.18×10 ⁻⁶	-0.247	4.58×10 ⁻²	-0.108
			T cell NK	3.18×10 ⁻⁶	-0.247	2.02×10 ⁻⁶	0.253
			Macrophage	3.18×10 ⁻⁶	-0.247	1.70×10 ⁻²	0.128
			B cell	3.18×10 ⁻⁶	-0.247	3.24×10 ⁻⁴	0.192
			Dendritic cell	3.18×10 ⁻⁶	-0.247	3.54×10 ⁻⁷	0.276
		MAPK13	CD4 ⁺	1.75×10 ⁻³	-0.168	2.06×10 ⁻¹⁶	0.423
			CD8 ⁺	-	-	-	Insignificant
			T cell NK	1.75×10 ⁻³	-0.168	1.22×10 ⁻⁶	0.258
			Macrophage	1.75×10 ⁻³	-0.168	2.94×10 ⁻¹²	0.364
			B cell	1.75×10 ⁻³	-0.168	1.33×10 ⁻¹²	0.37
			Dendritic cell	1.75×10 ⁻³	-0.168	8.64×10 ⁻²⁶	0.525
2.	LUAD	MAPK6	CD4 ⁺	4.31×10 ⁻²	0.525	6.71×10 ⁻³	-0.122
			CD8 ⁺	4.31×10 ⁻²	0.525	3.34×10 ⁻⁵	0.186
			T cell NK	4.31×10 ⁻²	0.525	4.50×10 ⁻¹⁹	-0.387
			Macrophage	4.31×10 ⁻²	0.525	3.44×10 ⁻¹³	0.32
			B cell	4.31×10 ⁻²	0.525	9.94×10 ⁻¹⁶	-0.351
			Dendritic cell	-	-	-	Insignificant

MAPK: Mitogen-activated protein kinase; LIHC: liver hepatocellular carcinoma; LUAD: lung adenocarcinoma.

Despite the limited number of cancer types showing statistical significance, the structural and functional interconnectivity of these MAPKs provides a rationale for co-targeting strategies. Hence, targeting up-regulated MAPKs, MAPK3, MAPK7, MAPK9, MAPK12, and MAPK13

for LIHC and MAPK6 for LUAD can be very crucial for therapeutic development of cancers. These findings not only align with prior evidence but also reinforce the potential of these MAPKs as promising therapeutic targets, particularly in LUAD and LIHC. Also, to complement our

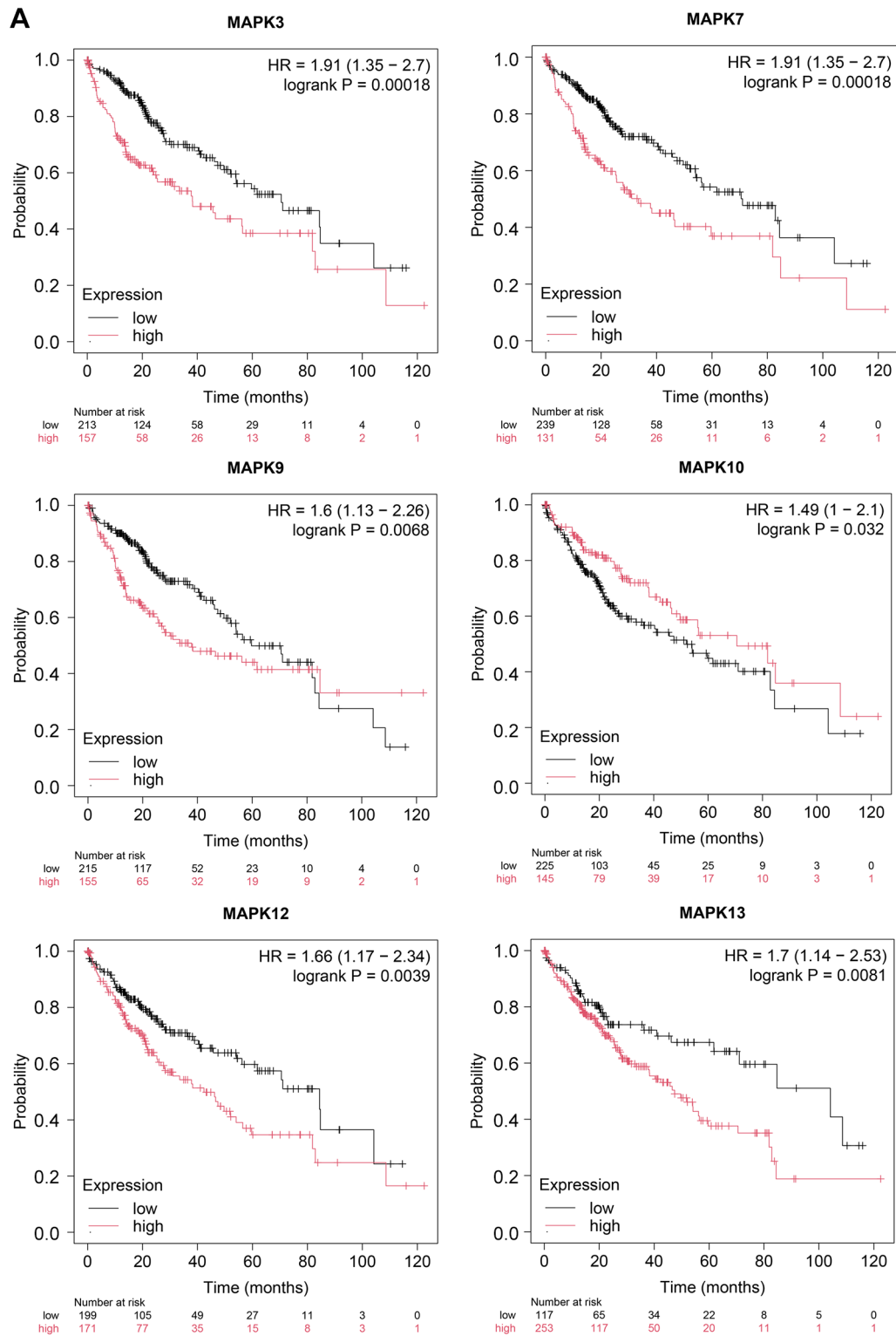


Figure 7. Continued

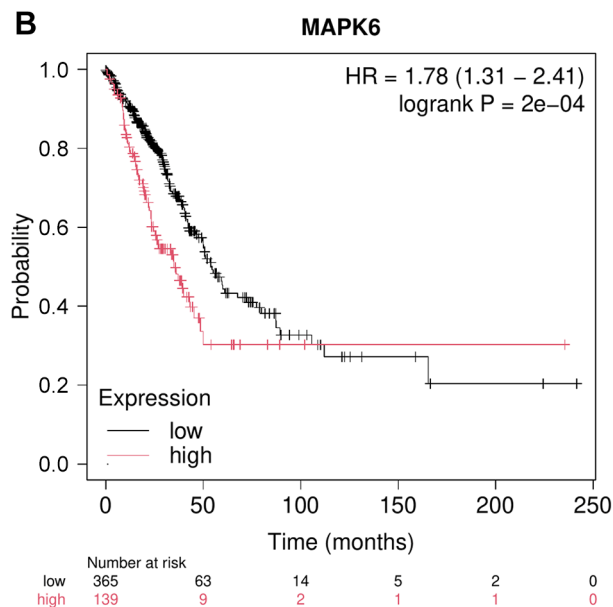


Figure 7. Overall Kaplan-Meier survival estimation of (A) LIHC patients and (B) LUAD patients.

transcript-level analyses, we reviewed independent protein-level studies demonstrating MAPK activation by quantitative proteomics and Western blot across multiple cancers. For example, Suzuki *et al.* (39) showed ERK1/2 suppression by acyclic retinoid in melanoma *via* Western blot, Dong *et al.* (40) reported irradiation-induced p-ERK elevation in breast and cervical cancer, and proteomic profiling of pilocytic astrocytoma confirmed ERK-2 overexpression by western blot (41). Although from other tumor types, these data consistently show that transcript-level MAPK alterations correspond to protein-level changes, reinforcing our findings. Together with our results, this evidence supports our elucidated MAPKs as potential biomarkers for LIHC and for LUAD, warranting future validation in patient tissues *via* western blot or immunohistochemistry.

Conclusion

This study provides a comprehensive, multi-layered analysis of MAPK gene family members across nine

prevalent human cancers, focusing particularly on their expression, mutational landscape, survival relevance, pathway associations, and immune infiltration patterns. Among the analyzed cancer types, LIHC and LUAD exhibited the most prominent MAPK dysregulation. Specifically, six MAPKs MAPK3, MAPK7, MAPK9, MAPK10, MAPK12, and MAPK13 were found to be significantly altered in LIHC, while MAPK6 showed a notable up-regulation in LUAD. These genes not only demonstrated functional enrichment in pathways central to tumor growth, immune evasion, and transcriptional regulation, but also correlated significantly with patient survival outcomes, indicating their potential as prognostic biomarkers. The tumor immune infiltration analysis further emphasized the role of these kinases in shaping the tumor microenvironment, which is crucial for developing effective immunotherapeutic strategies. Importantly, since MAPKs are structurally conserved and often function within interconnected signaling cascades, simultaneously targeting multiple family members could improve therapeutic outcomes and reduce resistance. Our findings advocate for the inclusion of MAPKs in the portfolio of precision oncology targets and support their development as biomarkers for early diagnosis, prognosis, and therapeutic intervention. Future studies involving *in vitro* validation and clinical trials are crucial to translate these findings into actionable cancer therapies. Overall, this integrative bioinformatics approach not only highlights the oncogenic relevance of MAPKs in LIHC and LUAD but also strengthens the rationale for family-wide targeting of MAPKs in cancer treatment strategies.

Conflicts of Interest

The Authors declare no conflicts of interest.

Authors' Contributions

Md Nayab Sulaimani: Conceptualization, investigation, methodology, writing—original draft preparation; Prithvi Singh: Data curation, graphics, data validation; Mohammad

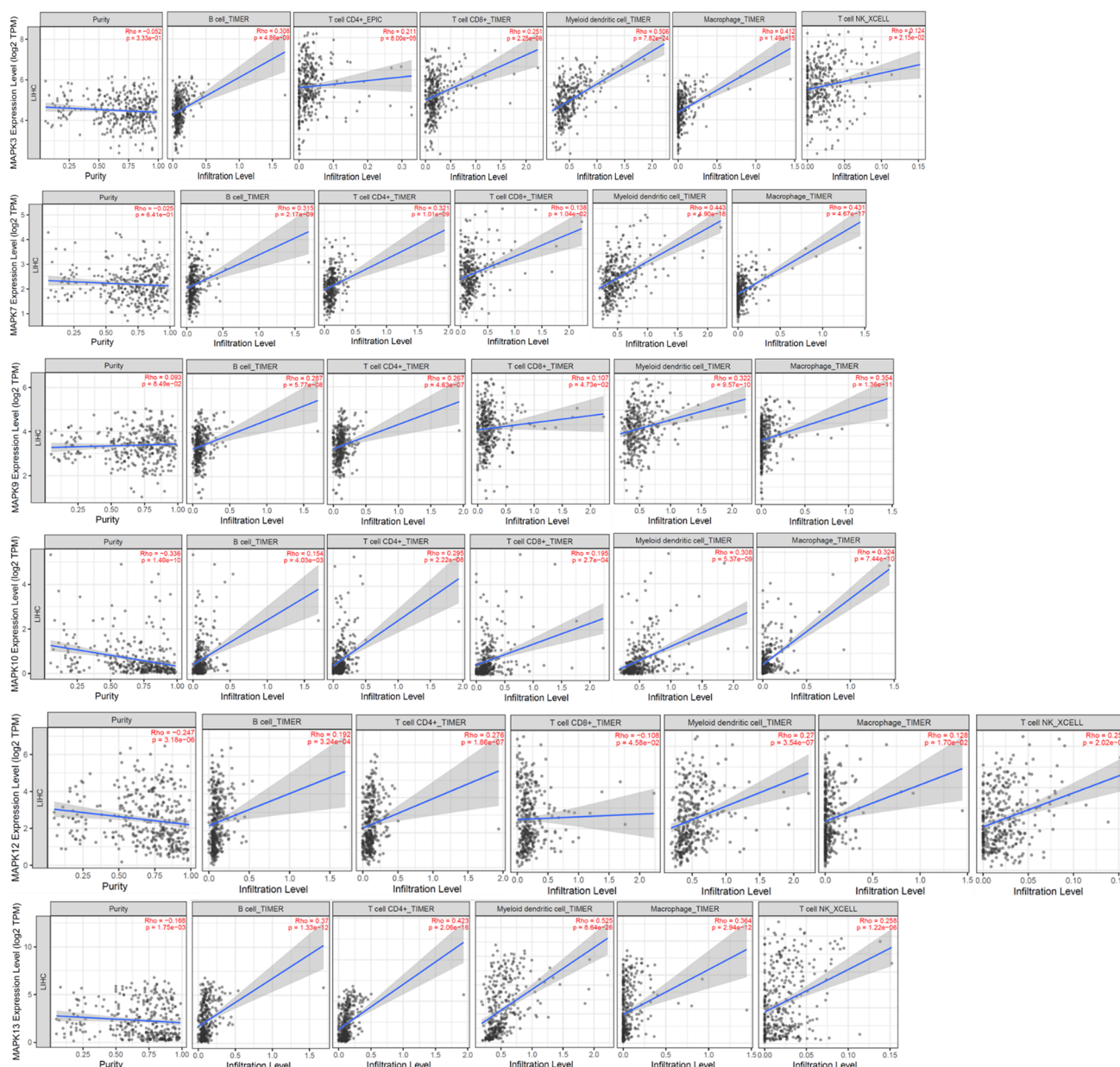


Figure 8. Scatterplots showing significant correlations of mitogen-activated protein kinase (MAPK) 3, MAPK7, MAPK9, MAPK10, MAPK12 and MAPK13 mRNA expression levels against tumor purity and various immune cell infiltration levels across The Cancer Genome Atlas (TCGA)-liver hepatocellular carcinoma (LIHC) cohort. mRNA expression levels against tumor purity are demonstrated on the left panel.

Umar Saeed: Methodology, data validation, writing-review, and editing; Sobia Zaidi: Data curation, visualization, methodology, writing-original draft; Afzal Hussain: Conceptualization, writing-review & editing, validation, supervision, investigation, funding acquisition;

Mohamed F. Alajmi: Investigation, data curation, methodology, writing-review & editing; Anas Shamsi: Data curation, graphics, data validation; Ravins Dohare: Methodology, writing-review, and editing, data validation, supervision; Md. Imtaiyaz Hassan: Conceptualization,

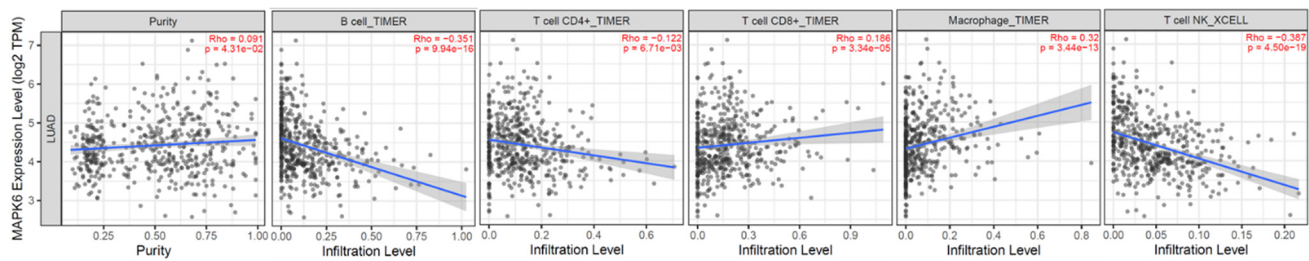


Figure 9. Scatterplots showing significant correlations of MAPK6 mRNA expression with $CD4^+$, $CD8^+$, T cell NK, macrophage, B cell, and dendritic cell infiltrating levels across The Cancer Genome Atlas (TCGA)-lung adenocarcinoma (LUAD) cohort. mRNA expression levels against tumor purity are demonstrated on the left panel.

investigation, supervision, review and editing, and project administration.

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Artificial Intelligence (AI) Disclosure

No artificial intelligence (AI) tools, including large language models or machine learning software, were used in the preparation, analysis, or presentation of this manuscript. Only ChatGPT, OpenAI was used for the refinement of language.

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